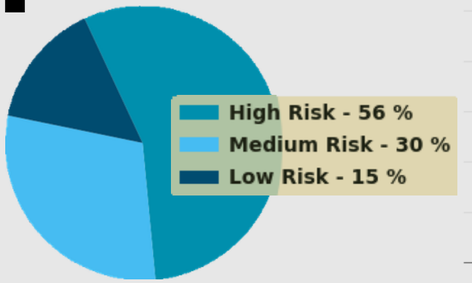
PROBLEM STATEMENT

India's banking ecosystem is seeing a surge in document fraud, identity manipulation, and synthetic submissions that move faster than manual verification can catch. Traditional OCR can extract text, but it cannot reliably detect visual inconsistencies, metadata tampering, or pixel-level edits.

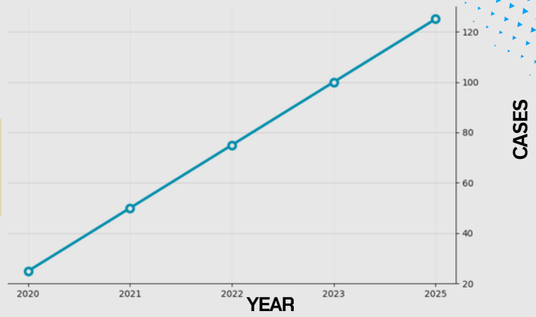
₹50,000+ Crore
TOTAL ESTIMATED NPA RISK

12 to 24 Days
AVERAGE MANUAL PROCESSING TIME

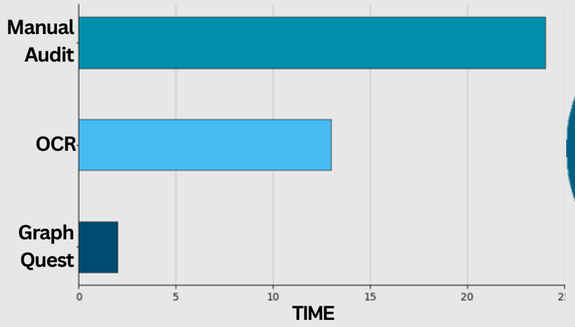
82 %
INTRUSIONS WITH NO VISUAL SIGNATURE



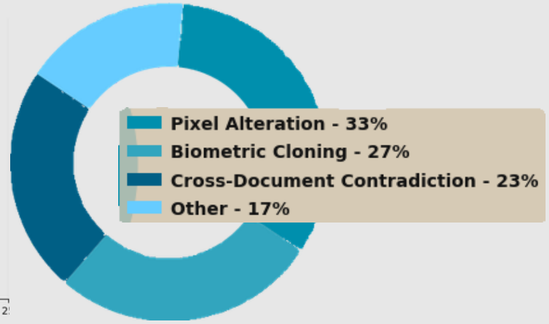
BANKING SECTOR VULNERABILITY



DOCUMENT FRAUD CASES



DETECTION BY METHOD



TYPES OF DOCUMENT FRAUD

KEY INSIGHTS

- ✓ Chennai Multi-Pledge Fraud Case: The same property was allegedly pledged across multiple banks due to lack of real-time mortgage visibility and cross-bank verification.
- ✓ Lucknow Synthetic Identity Fraud Case: Fraudsters reportedly used fake/deceased identity records and manipulated ownership details to attempt fraudulent loan approvals.
- ✓ Indore Forged Registry Case: Edited land records, forged registry papers, and manipulated metadata exposed weaknesses in traditional OCR-based verification systems.
- ✓ Most underwriting workflows still operate in silos, creating blind spots across banks, registries, and financial institutions.
- ✓ Cross-database validation and graph intelligence are essential for identifying hidden fraud relationships and repeat offenders.
- ✓ AI-powered forensic verification can significantly reduce underwriting delays, duplicate pledges, and financial exposure before loan disbursement.

PROPOSED SOLUTION

DhanRakshak — Autonomous Underwriting Intelligence & Fraud Prevention Platform

DhanRakshak is an end-to-end underwriting verification platform that helps banks detect forged documents, fake identities, duplicate property pledges, and ownership fraud before loan approval. The system combines digital forensics, OCR intelligence, government database validation, fraud correlation, and intelligent automation to transform manual underwriting into a faster, evidence-based verification workflow.

How Our Solution Works

DOCUMENT FORENSICS

- Borrower uploads KYC, land records, property documents, income proofs, bank statements etc.
- All documents are hashed, time-stamped and securely stored for auditability.

FORENSIC VISION ENGINE

- Performs pixel-level tamper detection using ELA, noise analysis, compression artifacts, and signature verification.
- Detects AI-generated or cloned documents, altered seals, forged signatures and metadata manipulation.

OCR & NLP INTELLIGENCE

- LayoutLM + Vernacular OCR extracts text from complex layouts and regional documents.
- Entities such as name, DOB, survey number, owner name, property ID. loan amount etc. are extracted and normalized.

AUTONOMOUS AI AGENT

- Our autonomous agent (built with MCP orchestration) reason over extracted entities.
- It decides what to verify, where to verify and how to verify.
- It triggers real-time API calls, compares responses and identifies contradictions.

GOVERNMENT VALIDATION LAYER

- AI agent verifies data against multiple trusted government sources:
  - Bhulekh (land records)
  - CERSAI (secured assets)
  - Municipal Corporation
  - Death Registry
  - Encumbrance Certificate

GRAPH INTELLIGENCE ENGINE

- All verified entities are stored in a knowledge graph.
- It maps relationships between people, properties, banks and documents.
- Detects patterns like duplicate pledges, common addresses, linked fraudsters, and circular ownership.

RISK & RESPONSE ENGINE

- Generates fraud confidence score (0-100).
- High-risk cases are auto-blocked.
- Alerts are sent to underwriting team/SOC.
- Complete audit trail + explainable reasoning is generated.

HOW DHANRAKSHAK FIXES UNDERWRITING LOOPHOLES

Real World Fraud Cases & Loopholes

CASE 1 : Multi - Pledge Fraud

Same property is used for loans in multiple banks.

? No Cross bank mortgage visibility



CASE 2 : Synthetic Identity Fraud

Fake/deceased identities used for loan approval

? No real world identity verification



CASE 3 : Forged Document/AI - Generated Registry

Forged land records, AI-generated documents, manipulated signatures

! OCR cannot detect tampering or edits



Case-1: Multi-pledge fraud detection

Loan application received

Borrower submits

- property documents
- sale deed
- KYC number

OCR + entity extraction

Dhanrakshak extracts

- owner name
- property ID
- Registry details

Owner Name	:	Ramesh Patil
Property ID	:	MH21AB1234
Survey No.	:	45/2/1
Registry No.	:	2018/56789

Graph intelligence check

Scans

- CERSAI records
- linked bank mortages
- historical bank loans

CROSS-BANK DUPLICATE DETECTED

Fraud match found

Same property already pledged at two other institutions



Automated decision

- Loan blocked
- fraud score generated
- alert sent

FRAUD PREVENTED BEFORE DISBURSEMENT

₹4.2 Cr Potential Fraud Prevented | 11 Seconds Verification Completed | 3 Banks Duplicate Pledge Detected

TECHNICAL ARCHITECTURE

MULTI SOURCE INPUT

1. **Loan Application Data** — Borrower & loan details
2. **KYC Documents** — Identity verification documents
3. **Property Documents** — Ownership & property records
4. **Bank Statements & Financial Documents** — Financial income verification
5. **Income / Tax Records** — Salary & tax validation
6. **External Databases & Government Records** — Legal & compliance verification

Intelligent Verification Engine

AI + Graph + Rules-Based Intelligence

The central fraud intelligence layer is envisioned to integrate order logic into the fraud intelligence arsenal and co-create secure digital evidence, tamper-aware document intake, and compliance-ready logging.

INGESTION LAYER
Main Function: Secure document upload & validation
Main Purpose: Create tamper-proof digital evidence

PROCESSING AND EXTRACTION LAYER
Main Function: OCR/NLP-based data extraction
Main Purpose: Convert documents into machine-readable data

VERIFICATION AND VALIDATION LAYER
Main Function: Cross-check data with databases and records
Main Purpose: Detect fraud and verify authenticity

INTELLEGENCE AND CORRELATION LAYER
Main Function: Correlate entities and analyze risk patterns
Main Purpose: Identify hidden fraud and generate risk insights

DECISION & RISK LAYER
Main Function: Risk scoring and decision support
Main Purpose: Assist human underwriters in final decision

Architecture Highlights

- **Microservices-Based Architecture** — Independent scalable modules
- **Cloud Agnostic** — AWS, Azure & on-premise deployment
- **AI-First Intelligence Layer** — Embedded verification intelligence
- **Scalable & Fault Tolerant** — High availability processing
- **Secure by Design** — Encryption & audit logs

System Outputs

- **Fraud Alert & Risk Score** — Real-time risk classification
- **Verification Report** — Evidence-based verification summary
- **Underwriter Dashboard** — Fraud insights & monitoring
- **Case Tracking & Audit Trail** — Full verification traceability
- **Regulatory & Compliance Reporting** — Audit-ready compliance reports

INTELLIGENCE FOUNDATION

- **Operational Database** — Workflows, transactions, verification events
- **Fraud Intelligence Graph Database** — Maps entities, relationships, fraud patterns
- **Document Vault** — Encrypted secure document storage
- **Audit Logs** — Immutable activity tracking
- **Audit Repository** — Immutable activity tracking
- **Model Repository** — Stores AI models, rules, fraud detection policies

BY THE NUMBERS (INDUSTRY CONTEXT)

- **₹1.25 Lakh+ Crore** — Estimated financial fraud losses in India
- **70+** — Frauds happen during loan underwriting & documentation
- **5–20 Days** — Average manual verification turnaround time
- **60–80%** — Frauds involve forged documents, fake IDs or duplicate pledges
- **53%** — Banks face challenges in real-time property verification
- **3X** — Rise in synthetic identity fraud cases

DATA FLOW OVERVIEW

1. **UPLOAD DOCUMENTS** — Scanned fon with upload documents
2. **EXTRACT INFORMATION** — Extract scnoam documents
3. **VALIDATE WITH AUTHORITIES** — Embedded with official seal
4. **CORRELATED & DETECT ANOMALIES** — Fraud insights & detect anomalies
5. **GENERATE RISK SCORE** — Evidence-based nisker summary
6. **ALERT / APPROVE / ESCALATE** — Alert / approve / comprate geel

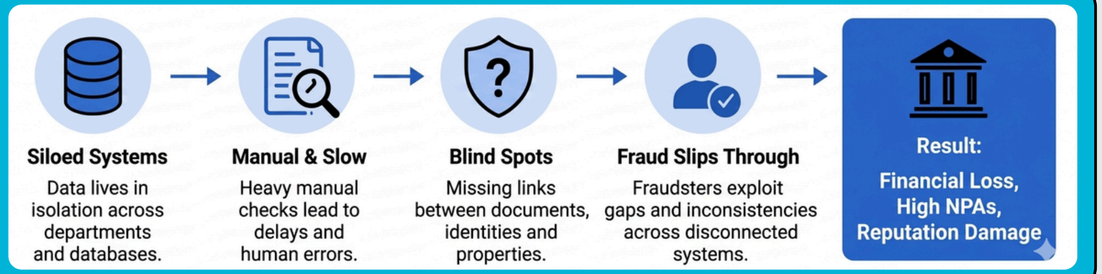
THE MARKET HAS TOOLS.
WE BRING THEM TOGETHER FOR UNDERWRITING.

How Existing Solutions Compare

Capability	Doc Forensics	ID/KYC Verification	Property/ Gov Validation	Cross-Entity Correlation	Underwriting Focus
OCR/ Scanning Tools	✓ (Basic)	✗	✗	✗	✗
KYC/ID Platforms	✗	✓	✗	✗	✗
Fraud Analytics/ Graph Platforms	✗	✗	–	✓	✗
Mortgage /Property Data Providers	–	–	✓ (Limited)	✗	✗
Our Platform (Unified)	✓	✓	✓	✓	✓

✓ Full capability
✗ Partial/Limited
– Not available

THE PROBLEM WITH TODAY'S APPROACH



OUR DIFFERENTIATION



UNDERWRITING FRAUD: THE BIG PICTURE



BEYOND VERIFICATION

The Future of Intelligent Underwriting Security

Building a Connected Fraud Intelligence Ecosystem for Safer Financial Decisions

FUTURE VISION

Turning fragmented verification into a connected national fraud intelligence ecosystem.

- AI-assisted underwriting intelligence
- Cross-bank fraud correlation
- Scalable verification workflows
- Explainable fraud decision support
- Secure, compliant & regulator-ready

RESEARCH AND REFERENCES

- RBI Banking Fraud Reports
- CERSAI Documentation
- Bhulekh Systems
- Deloitte Fraud Insights
- KPMG Banking Risk Reports
- Neo4j Fraud Intelligence Studies
- OCR & Document Forensics Research Papers

KEY IMPACT

70%+
Reduction in manual underwriting effort

<5 sec
Future verification response target

Nationwide
Cross-bank fraud intelligence capability

AI-assisted
Explainable decision workflows

Scalable
Enterprise deployment across banks